

Bulk Polygon Registration using GeoJSON

This document outlines the process for registering multiple polygons at once using a GeoJSON file, specifically designed to handle up to 1000 polygons in a single request. This method is preferred when dealing with a high volume of polygons, as the lengthy response messages can be difficult to display and manage in tools like Swagger.

Purpose

The primary purpose of this functionality is to provide an efficient and reliable way to register a large number of geographical polygons (up to 1000) through a single API call, minimizing network overhead and improving the user experience compared to registering them individually.

GeoJSON Format for Input

The input payload must be a single GeoJSON FeatureCollection object. Each feature within the collection must be a **Polygon** geometry type.

Example GeoJSON Structure

The GeoJSON file should adhere to the following structure:

```
{  
  "type": "FeatureCollection",  
  "features": [  
    {  
      "type": "Feature",  
      "properties": {  
        "s2_index": "8,13"  
      },  
      "geometry": {  
        "type": "Polygon",  
        "coordinates": [  
          [  
            75.782,
```

```

        30.907
      ],
      [
        75.7828,
        30.907
      ],
      [
        75.7828,
        30.9078
      ],
      [
        75.782,
        30.9078
      ],
      [
        75.782,
        30.907
      ]
    ]
  ]
}
},
{
  "type": "Feature",
  "properties": {
    "s2_index": "8,13"
  },
  "geometry": {
    "type": "Polygon",
    "coordinates": [
      [
        [
          75.783,
          30.907
        ],
        [
          75.7838,
          30.907
        ],
        [

```

```
    75.7838,  
    30.9078  
  ],  
  [  
    75.783,  
    30.9078  
  ],  
  [  
    75.783,  
    30.907  
  ]  
]  
}  
}  
]
```

Constraints and Best Practices

Constraint	Description
Maximum Limit	The request is restricted to a maximum of 1000 polygons per GeoJSON file. Requests exceeding this limit will be rejected.
Geometry Type	Only the Polygon geometry type is accepted for feature registration.
Coordinates	Polygons must be properly closed (the first and last coordinate pair must be identical).
Payload Size	Large GeoJSON files may impact performance. Ensure the file size is reasonable for transmission.

Step-by-Step Registration Process

The following steps guide you through authenticating your session and submitting your GeoJSON file for bulk polygon registration using the provided API documentation.

Swagger UI Link : <https://api-ar.agstack.org/swagger/>.

1. Obtain Your Access Token

Before making requests, you must generate a JSON Web Token (JWT) using your account credentials:

- Navigate to the **Authentication** section.
- Select the **POST /login** endpoint and click **Try it out**.
- Enter your credentials in the request body:

```
{  
  
  "email": "your_email@gmail.com",  
  
  "password": "your_password"  
}
```

- Click **Execute**.
- From the server response, copy the string value provided for the **"access_token"**.

2. Authorize Your Session

Once you have the token, you must apply it to the Swagger interface to unlock protected endpoints:

- Scroll to the top of the page and click the **Authorize** button (top right).
- In the "Value" field, paste your copied **access_token**.
- Click **Authorize**, then click **Close**.

Note: You are now authenticated. A small "locked" icon should appear next to the API endpoints.

3. Register Field Boundaries (GeoJSON)

With your session authorized, you can now upload your spatial data:

- Locate the **Field Registration (POST)** section.
- Select the `/register-field-boundaries-geojson` API endpoint.
- Click **Try it out**.
- Click **Choose File** to upload your `.geojson` file containing the polygons you wish to register.
- Click **Execute**.

The system will process the file and return a success message confirming your polygons are registered.